

A Higher-Smoothness Upgrade for the L_p Branch of the $c_0(\Gamma)$ Embedding

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Status

This packet records a likely valid partial answer to a question in the 2024 paper of Hajek, Johanis, and Schlumprecht on point character and embeddings into $c_0(\Gamma)$ [1]. Their Corollary 28 gives, for subspaces of $L_p(\mu)$ with $1 < p < \infty$ and density below ω_ω , a bi-Lipschitz embedding

$$\Phi : X \longrightarrow c_0(\Gamma)$$

whose scalar coordinate functions are C^1 smooth. Immediately after the corollary, they ask whether for $p \geq 2$ the component functions can have higher smoothness.

The answer is affirmative for the L_p -subspace branch in the natural range allowed by the smoothness of the L_p norm: C^m for every integer $m < p$, and for every finite m when p is an even integer.

Source Question

The source passage is on page 11 of arXiv:2401.00831.

the following corollary (note that any Lipschitz function from a subset of a metric space can be extended to the whole space with the same Lipschitz constant):

Corollary 28. *Let X be a subspace of $L_p(\mu)$ for some measure μ and $1 < p < \infty$, resp. of some super-reflexive Banach lattice with a (long) unconditional basis or a weak unit, with $\text{dens } X < \omega_\omega$. Then there is a bi-Lipschitz embedding $\Phi : X \rightarrow c_0(\Gamma)$ such that the component functions $f_\gamma \circ \Phi$ are C^1 -smooth, where f_γ are the canonical coordinate functionals on $c_0(\Gamma)$.*

We do not know whether in the corollary above in the case $p \geq 2$ the component functions can be of higher smoothness. Invoking [\[HJ\]](#) Theorems 7.79 and 7.86] we obtain a corollary on approximation of Lipschitz mappings:

Corollary 29. *Let X be a subspace of $L_p(\mu)$ for some measure μ and $1 < p < \infty$, resp. of some super-reflexive Banach lattice with a (long) unconditional basis or*

In text form, after proving the C^1 -coordinate embedding corollary, the authors write:

We do not know whether in the corollary above in the case $p \geq 2$ the component functions can be of higher smoothness.

Background: Property $(*^m)$

Definition 1. Let $m \in \mathbb{N}$. A Banach space X has property $(*^m)$ if there is a constant $C_0 \geq 1$ such that for every Lipschitz function $f : X \rightarrow \mathbb{R}$ and every $\varepsilon > 0$, there is a C^m -smooth Lipschitz function $g : X \rightarrow \mathbb{R}$ with

$$|f(x) - g(x)| < \varepsilon \quad (x \in X), \quad \text{Lip}(g) \leq C_0 \text{Lip}(f).$$

This is the standard Lipschitz smooth-approximation property used in the Hajek–Johanis smooth approximation framework. Fry and Keener prove that a WCG Banach space with a C^m smooth norm has property $(*^m)$ [3, Theorem 4]. Their result is stated for C^p smoothness; for an integer m we apply it with $p = m$.

Proof Intuition

The $c_0(\Gamma)$ -valued part of the source proof is metric: once $\text{dens } X < \omega_\omega$, the paper gives a bi-Lipschitz embedding of X into some $c_0(\Gamma)$. Smoothness of the scalar coordinates is supplied afterwards by a separate coordinate-smoothing theorem of Hajek and Johanis.

That smoothing theorem does not use a specifically first-order argument. Its input is the ability to approximate scalar Lipschitz functions by smooth Lipschitz functions with controlled Lipschitz constants; the support and threshold bookkeeping that keeps the map c_0 -valued and bi-Lipschitz is independent of the differentiability order. Therefore replacing property $(*^1)$ by $(*^m)$ replaces C^1 coordinates by C^m coordinates.

For L_p -subspaces, the remaining input is exactly the classical smoothness of the L_p norm together with Fry–Keener’s WCG approximation theorem.

Order- m Coordinate Smoothing

Lemma 1 (Order- m Hajek–Johanis coordinate smoothing). Let $m \in \mathbb{N}$. Suppose a Banach space X has property $(*^m)$. If X admits a bi-Lipschitz embedding into some $c_0(\Gamma)$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose coordinate functions are C^m smooth.

Proof. This is the order- m form of the coordinate-smoothing theorem used by Hajek–Johanis–Schlumprecht through [2, Theorem 7.63] in their proof of the C^1 -coordinate corollary. We spell out the only point that depends on differentiability order.

In the Hajek–Johanis construction, a bi-Lipschitz $c_0(\Gamma)$ -embedding is first converted into a family of real-valued Lipschitz coordinate tests with fixed quantitative margins. These tests are arranged in point-finite or thresholded families, so that at every $x \in X$ the resulting coordinate vector is in c_0 , and for every pair x, y at least one coordinate separates them by a fixed multiple of $\|x - y\|$. The last step replaces each real-valued Lipschitz test by a smooth Lipschitz approximant whose uniform error is chosen smaller than the relevant margin. The upper Lipschitz bound follows from the common Lipschitz control of the approximants; the lower Lipschitz bound survives because the approximation errors are below the separation margins; and the c_0 -valued property survives because the same point-finite/threshold support bookkeeping is used after the replacement.

For the C^1 statement, this last replacement uses property $(*^1)$. If X has property $(*^m)$, the identical replacement produces C^m -smooth approximants with the same Lipschitz and error estimates. No subsequent step differentiates the approximants or uses a feature special to $m = 1$. Thus every coordinate in the final $c_0(\Gamma')$ -valued map is C^m , and the same constants give a bi-Lipschitz embedding. $\square$

Main Partial Result

Theorem 1. *Let X be a closed subspace of $L_p(\mu)$, where $2 \leq p < \infty$, and suppose that $\text{dens } X < \omega_\omega$. Let $m \in \mathbb{N}$.*

If $m < p$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose coordinate functions are C^m smooth. If p is an even integer, the same conclusion holds for every finite m .

Proof. By Corollary 26 of [1], X admits a bi-Lipschitz embedding into some $c_0(\Gamma)$, because $X \subset L_p(\mu)$ and $\text{dens } X < \omega_\omega$.

Since $1 < p < \infty$, the space $L_p(\mu)$ is reflexive, and every closed subspace X is reflexive. In particular X is weakly compactly generated. For every integer $m < p$, the usual L_p -norm is C^m smooth away from the origin; when p is an even integer, the corresponding power function is polynomial and the usual smooth-norm argument gives C^m smoothness for every finite m . Restricting the norm to X therefore gives X a C^m smooth norm in the stated range.

Fry–Keener [3, Theorem 4] now gives property $(*^m)$ for X . Applying Lemma 1 to the bi-Lipschitz $c_0(\Gamma)$ -embedding supplied by the source paper yields a bi-Lipschitz embedding of X into another c_0 -space with C^m smooth coordinate functions. $\square$

Corollary 1. *For every closed subspace $X \subset L_p(\mu)$ with $2 \leq p < \infty$ and $\text{dens } X < \omega_\omega$, the embedding in the source corollary can be chosen with coordinate functions smoother than C^1 . More precisely, one can take all coordinates C^2 when $p > 2$, all finite orders when $p = 2$, and all orders $m < p$ in general.*

Proof. If $p = 2$, apply Theorem 1 in the even-integer case. If $p > 2$, take $m = 2$, or any larger integer $m < p$ when available. $\square$

Verification Notes

The proof uses three external inputs.

First, the source paper supplies the underlying bi-Lipschitz embedding of L_p -subspaces of density below ω_ω into $c_0(\Gamma)$ [1, Corollary 26]. Second, Hajek–Johanis coordinate smoothing is used in the same way the source uses it for the C^1 corollary; the only change is replacing property $(*^1)$ by property $(*^m)$. Third, Fry–Keener supplies property $(*^m)$ for WCG spaces with a C^m smooth norm [3, Theorem 4].

No numerical or symbolic computation is involved.

Limitations

This is a partial answer to the source question. It covers the L_p branch of the source corollary in the differentiability range supported by the L_p norm. It does not settle the super-reflexive Banach-lattice branch unless the lattice in question is separately known to satisfy property $(*^m)$. It also does not claim C^m coordinates for non-even L_p -spaces in orders $m \geq p$.

Novelty Check

The run indexes were searched for arXiv:2401.00831, smooth-coordinate keywords, C^m /higher-smoothness phrases, Fry–Keener, point character, and $c_0(\Gamma)$ embedding terminology. The only prior run item was the conditional packet isolating this same route. A bounded web search for

the source title together with “higher smoothness”, “component functions”, and “Theorem 7.63” did not reveal a later explicit answer. The present packet therefore records the route as new run progress, subject to human review of the order- m reading of the Hajek–Johanis coordinate-smoothing theorem.

Human Review Recommendation

Review Lemma 1 first. The substantive check is that the Hajek–Johanis coordinate-smoothing theorem used by the source is indeed order-free once property $(*^m)$ is available. After that, the Fry–Keener and L_p -smoothness inputs are standard.

References

- [1] P. Hajek, M. Johanis, and T. Schlumprecht, *Remarks on the point character of Banach spaces and non-linear embeddings into $c_0(\Gamma)$* , arXiv:2401.00831, 2024.
- [2] P. Hajek and M. Johanis, *Smooth analysis in Banach spaces*, De Gruyter Series in Nonlinear Analysis and Applications, vol. 19, De Gruyter, Berlin, 2014.
- [3] R. Fry and L. Keener, *Approximation of Lipschitz functions by Lipschitz, C^p smooth functions on weakly compactly generated Banach spaces*, arXiv:0907.0241, 2009.